01 / PROBLEM AND CONTRIBUTION

# GCN accelerator

EEE 525 · VLSI Design | Spring 2026 | Academic design

**RESULT AT A GLANCE**

Reproduced reference classifications at three implementation stages-RTL, post-synthesis, and post-route-using the same testbench through the ASIC flow.

## The engineering question

How can graph-node classification be mapped into a controlled RTL datapath and carried through an ASIC design flow?

## Situation

Graph-node classification combines dense feature transformation, sparse neighbor aggregation, and class selection. The project maps those phases into RTL and follows the design through physical implementation.

## Task

Coordinate the datapath with an FSM, reproduce the reference classifications, and carry the design through synthesis and place-and-route.

## My contribution

GCN datapath and controller design, simulation, synthesis, and physical implementation documented in the lab report.

## Tools and concepts

Verilog / RTL / FSM design / Design Compiler / Cadence Innovus / Static timing analysis

02 / DESIGN AND METHOD

# From the task to implementation

### 01 / Implementation step

Separated control and computation with a central FSM for feature loading, weight loading, transformation, aggregation, and classification.

### 02 / Implementation step

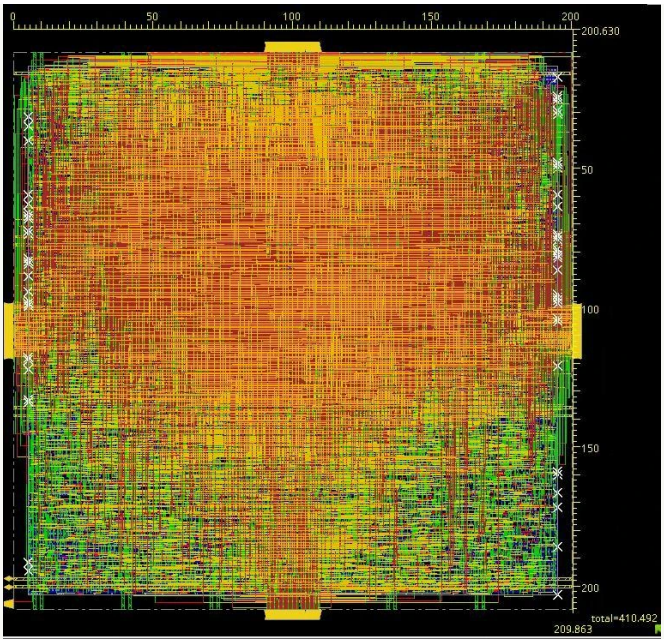
Used local weight storage and intermediate matrix buffers. The transformation stage computes feature-weight dot products; the aggregation stage traverses COO-format graph edges.

### 03 / Implementation step

Selected the largest of three output features for each node, then carried the RTL through Design Compiler synthesis and Innovus placement and routing.

### Dense and sparse stages

Feature transformation computes feature-weight dot products; aggregation follows the sparse graph edge representation. Local storage and intermediate buffers connect these stages under the controller rather than treating the algorithm as one undifferentiated operation.




**Routed accelerator layout**

Cadence Innovus implementation view from the GCN report.

03 / RESULTS AND ENGINEERING PERSPECTIVE

# Evidence, interpretation, and next steps

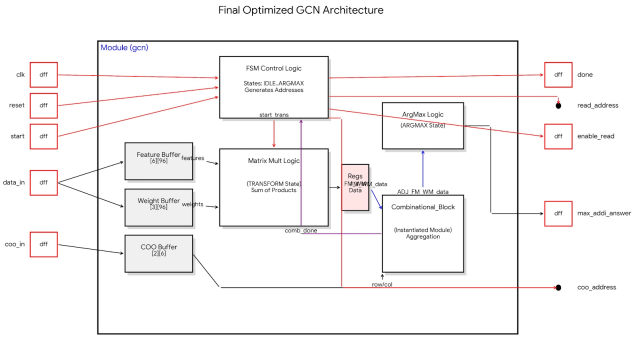
| | |
|---|---|
| **Validation stages** | 3 |
| **Class outputs** | 3 |

## Reported result

The same testbench reproduced the reference classifications at RTL, post-synthesis, and post-route stages. Setup and hold timing were examined after routing.

## Implementation continuity

The same reference-classification checks are used through RTL, synthesized, and routed design stages. This supports functional continuity for the tested data; examining timing reports is a separate implementation activity.




**Datapath and controller**

Reported GCN accelerator architecture.

## Scope and limitations

Matching the reference classifications across implementation stages is the main result. The design also illustrates the separation of dense feature transformation from sparse graph aggregation.

### Possible next steps - proposed, not completed results

- Add a reproducible input/reference-output example and tool invocation notes.
- Publish final timing constraints and unambiguous timing/area summaries before quoting additional PPA metrics.

Source material: EEE 525 Lab 4 milestone report, pp. 1-6, 8, 11-12; approved GCN brief.

Prepared from the supplied reports, project files, and verified portfolio copy. Figures retain their original content; explanatory diagrams are labelled.